The texwright visual cookbook

Working mathematics, diagrams, and cross-references

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1 Mathematics with a reference

For the illustrative sequence $a_n = 2^{-n}$, the first N terms satisfy

$$\sum_{n=1}^{N} 2^{-n} = 1 - 2^{-N}. \tag{1}$$

Equation (1) approaches one as N increases. References remain correct if another numbered equation is inserted earlier.

Table 1: Exact values from the geometric series, rounded to four places.

Terms N	Partial sum	Remaining gap
1	0.5000	0.5000
2	0.7500	0.2500
3	0.8750	0.1250
4	0.9375	0.0625

1.1 Plot the same relationship

The plot in Figure 1 is authored as TikZ and pgfplots text. There is no external chart image or TeX process.

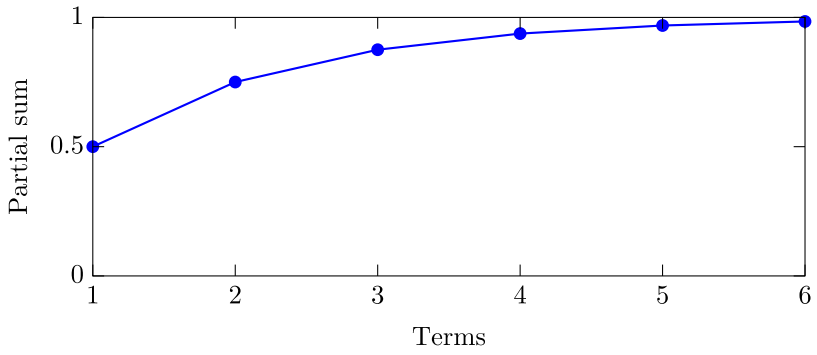


Figure 1: Partial sums approach one.

These numbers are mathematical examples, not performance measurements. Open this document's Markdown source to copy the formula, table, and plot.

2 Diagrams from readable source

A diagram can live beside the prose that explains it. Figure 2 describes the calls an agent makes when producing a document.

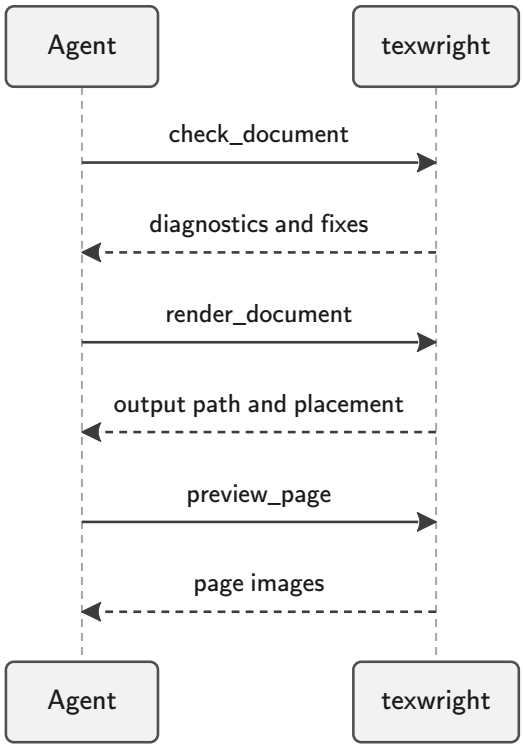


Figure 2: An agent requests a check, a render, and a page preview.

2.1 A useful delivery checklist

1. Keep the Markdown or LaTeX source with the PDF.
2. Include the referenced figures and bibliography.
3. Check diagnostics after the final edit.
4. Inspect the pages containing tables, mathematics, and figures.
5. Deliver an HTML version when searchable web text is useful.

> **Tip**
> This cookbook is its own example. Run the commands below from a built repository to reproduce the PDF and inspect page two.

```
node dist/cli.mjs check docs/cookbook.md --diagnostics compact
node dist/cli.mjs docs/cookbook.md -o docs/cookbook.pdf
node dist/cli.mjs preview docs/cookbook.md --page 2
```

For citations and theorem environments, continue with the research showcase. For a larger drawing collection, see the diagram examples.

All three guides, their HTML editions, reference schemas, and selected page images can be regenerated together with npm run docs.